

Hein Zarni Aung

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EDUCATION

Stony Brook University

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Software Security, Systems Fundamentals I & II, Algorithms, Full Stack Development

Expected Graduation: May 2027

RESEARCH

Undergraduate Research Assistant — Ethos Lab, SBU

PI: Abisheka Pitumpe | Team size: 2

Sep 2025 – Dec 2025 (paused)

- Built a **Python-based backend pipeline** to monitor newly issued SSL/TLS certificates in real time, scraping and analyzing HTML, CSS, JS, and URL data to flag potentially fraudulent domains impersonating EZ-Pass tolling services.
- Used Selenium to automate certificate/domain scraping and Python to filter and triage flagged sites; auto-captured homepage screenshots of promising candidates to speed up manual review.

INTERNSHIPS

Software Engineering Intern — Clustr

Cross-platform React Native social/scheduling app | ~8-person dev team

Jan 2026 – Sept 2026 (expected)

- Contributed to Clustr, a cross-platform **React Native** social/scheduling app, delivering 50+ commits across backend and frontend features.
- Built server-side backend infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows and activity feed endpoints.
- Designed and implemented a **React Native** Scheduler UI module featuring tabbed navigation, a reschedule bottom sheet, and custom date/time picker integration to improve user engagement.
- Collaborated with an 8-person, multi-developer team via Git, regularly merging and resolving conflicts across feature branches to keep modules in sync with the shared codebase.

Software Engineering Intern — Prenora Dynamics LLC

~8-person team

Jun 2026 – Sept 2026 (expected)

- Own end-to-end **backend design decisions** for an in-house, in-platform video chat feature, researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).

CAMPUS INVOLVEMENT

International Peer Mentor — Office of Global Affairs

Aug 2025 – Present

- Mentored 6 international students through the university's International Student Success peer mentoring program via biweekly one-on-one meetings on academic and cultural adjustment.

PROJECTS

TuneAcademy — Hack Brooklyn 2026 (36-hour hackathon)

Winner: Best Prenora Dynamics Hack (Sponsor Challenge) | Team of 4

Apr 2026

- Built the AI/audio-analysis pipeline for an app matching music students with instructors based on technical weaknesses, using Essentia and NumPy to extract pitch, rhythm, and tone features from recorded performances.
- Pivoted from a planned machine learning model to rule-based scoring algorithms under hackathon time/compute constraints, delivering reliable weakness detection within the 36-hour build window.
- Collaborated in a 4-person team; deployed the broader app's Python/FastAPI backend in Docker on **GCP hosting** with Cloud Firestore, alongside a React/Tailwind frontend and ElevenLabs-generated voice feedback.

Discord Voice Channel Activity Monitor

Node.js, Discord.js, Cloud

- Built a **Node.js**/Discord.js bot notifying users in real time when someone joins a voice channel, solving a scheduling problem for a friend group spread across different schedules.
- Deployed on a cloud server using PM2, reaching 50 concurrent active users; implemented customizable DM notifications and per-server/per-user configuration.

SKILLS

Languages: Python, JavaScript, TypeScript, Java, C

Backend: Node.js, Express.js, FastAPI

Frontend: React, React Native, Next.js

Databases & Cloud: MongoDB, PostgreSQL, Firebase/Firestore, AWS, GCP

Tools: Git, Docker, Linux, PM2, Selenium